

Unmasking Subtle Bias

A Two-Level Dual-View Event Relation Graph



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The Problem: Subtle Bias

Two Types of Bias Sentences:

S5 (Overt - Easy to Detect):

"Mr. Trump drew considerable criticism from the Republican establishment and attracted the ridicule from the public"

→ Obvious ideological language and emotionally loaded words (criticism, ridicule)

S6 (Subtle - Hard to Detect):

"Before this statement, Mr. Trump has run for president several times since the 1980s but has always decided not to follow through"

→ Looks completely neutral and factual

Why S6 is Actually Biased:

- Appears to report historical facts objectively
- BUT: The **placement** matters - it comes AFTER Trump's explanation for dropping out
- Event graph reveals: historical pattern UNDERMINES current claim
- Creates implicit skepticism about Trump's stated reason

The Challenge: Models examining sentences in isolation miss S6's bias. Need to understand event relationships shown in graph (right side).

Example Article		Event Relation Graph	
S1	Donald J. Trump said in a statement on Tuesday that he dropped plans to moderate a presidential debate.	statement	dropped
S2	Trump said that he dropped because the Republican Party asked him to give up the right to run as an independent candidate if he moderates the debate.	statement	dropped
S3	"It is very important to me that the right Republican candidate be chosen to defeat the failed Obama administration," Mr. Trump said in his statement .	statement	dropped
S4	"But if that Republican, in my opinion, is not the right candidate, I am not willing to give up my right to run as an independent candidate."	statement	dropped
S5	Mr. Trump drew considerable criticism from the Republican establishment and attracted the ridicule from the public for his decision to drop and his willingness to still run if he was displeased with the candidates.	statement	dropped
S6	Before this statement, Mr. Trump has run for president several times since the 1980s but has always decided not to follow through – he chose to drop the debate every time.	run	drop

statement

dropped

statement

dropped

run

drop

give up

give up

coreference

coreference

caused by

after

after

coreference

temporal

causal

subevent

The Research Gap

The Gap (State-of-the-Art):

- The best current method (Lei & Huang, 2024) uses an event graph, but it's **flat and unified**.
- It treats all events(facts and opinions) the same. This "flattens" the very signal we're looking for.

Our Key Insight:

Subtle bias is not in the events themselves, but in the **juxtaposition** of two distinct event types.

1. **Factual Events**
 - a. Objectively verifiable ("voted," "signed," "increased by 10%")
2. **Interpretive Events**
 - a. Subjective characterizations ("slammed," "warned," "could harm")

Our Hypothesis: To detect subtle bias, a model must **explicitly separate and model the interaction between facts and their interpretive framing**.

Our Solution: The Two-Level Dual-View Graph

Dual-View: separate the graph into two "views"—

- a. **Factual Subgraph:** The objective backbone of what happened
- b. **Interpretive Subgraph:** The subjective layer of how it's framed
- c. **Cross-View Edges:** Explicitly models the "fact-framing interaction"

Two-Level Hierarchy: We model this interaction at multiple scales (based on advisor feedback)—

- a. **Level 1**
 - i. **Paragraph Graph:** Captures local framing patterns
- b. **Level 2**
 - i. **Document Graph:** Captures global narrative structure and cross-paragraph undermining

One Fact, Two Opposite Interpretations

Why F/I Separation Matters

Factual Sentence (neutral)

S1: "The government announced a new 4% tax cut for middle-income workers."

- Factual events: announced, tax cut
- (No opinion, verifiable report)

Positive Interpretation (supportive framing)

S2: "Supporters say the cut will boost spending and help struggling families."

- Interpretive events: boost, help
- Cross-view alignment: boost → tax cut, help → tax cut
- Framing: beneficial, growth-oriented

Negative Interpretation (critical framing)

S3: "Critics argue the cut will drain public services and worsen inequality."

- Interpretive events: drain, worsen
- Cross-view alignment: drain → tax cut, worsen → tax cut
- Framing: harmful, regressive

Key Points:

- The factual event ("tax cut") is the same but the interpretations change
- The bias here is not in the fact; it's in the framing attached to it

Baseline model

sees all words mixed → cannot tell which ones are facts vs judgments

Dual-view model

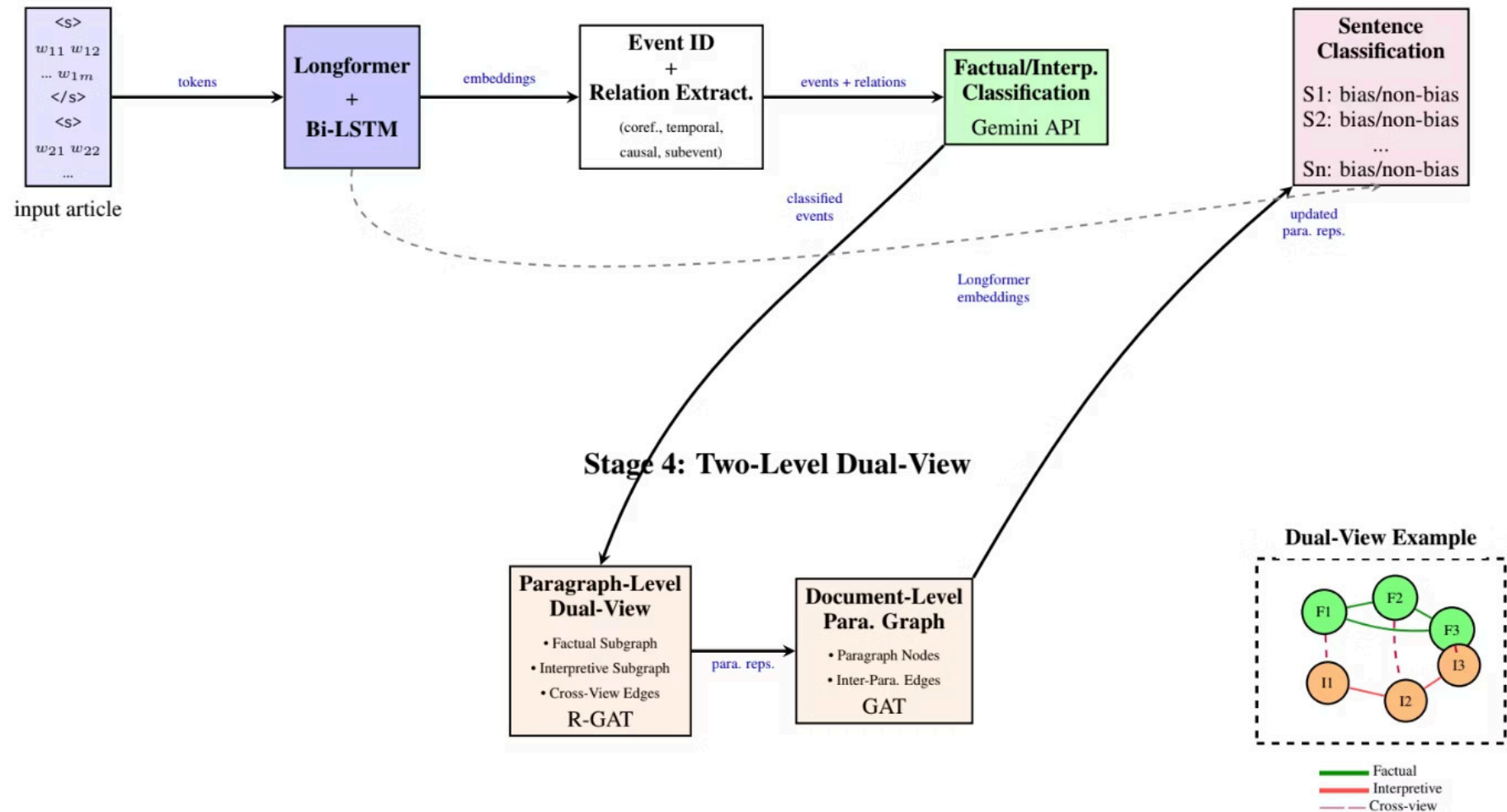
separates fact layer + interpretation layer → learns which framing is applied

Our 5-Stage Architecture

Stage 1: Encoding

Stage 2-3: Event Extraction

Stage 5: Classification



Stage 1-3: Graph Construction (Data Pipeline)

1. **Encode:** Input articles with Longformer + BiLSTM
2. **Extract:** Train on MAVEN-ERE to find all events and their 4 relations (Causal, Temporal, Coref, Subevent)
3. **Classify:** Use Gemini API to label all ~5,500 events as Factual (65%) or Interpretive (35%)

Implicit Relation Enhancement

The Problem:

News writing hides relations to appear neutral

Explicit (Easy):

"Failed because no funding"

Implicit (Hard):

"Failed. No funding."

Same relationship, no marker

Our Solution: 3 Techniques



1. Data Augmentation

- Remove markers from 20% of training data
- Remove: "because", "before", "includes"
- Forces model to learn structural patterns



2. Contrastive Learning

- Cluster similar relations together
- CAUSAL pairs → similar embeddings
- TEMPORAL pairs → different from CAUSAL



3. Enhanced Feature Layers

- Additional 512→256 layers per relation
- Extra capacity for subtle patterns

Why This Helps Bias Detection

Better Relations → Better Bias Detection

What We Can Now Extract:

✓ Implicit Temporal:

"Trump announced plans. He had abandoned similar ideas."

✓ Implicit Causal:

"Funding dried up. Program collapsed."

✓ Implicit Subevent:

"Conference featured speeches, panels, networking."

Impact:

- Richer event graphs
- More complete relation coverage
- Better detection of implicit framing patterns

Key: Real news avoids explicit markers → we must handle implicit relations

Stage 4-5: Bias Classification (Our Innovation)

1. **Process:** Run the graphs through our Two-Level Dual-View GNN
2. **Classify:** Make the final bias/non-bias sentence prediction

How Our Model "Thinks": A 3-Step Process

Step 1: Local (Paragraph) Processing	Step 2: Global (Document) Processing	Step 3: Final Classification
For each paragraph, we split the graph into two "views": Factual (F) and Interpretive (I)	We treat each paragraph summary (from Step 1) as a single "super-node" in a new document graph.	To make a final bias/non-bias prediction for a sentence, we combine three powerful signals
A Relation-Aware GNN (R-GAT) processes each view separately	Paragraphs are connected by adjacency (what comes next) and coreference chains (which paragraphs talk about the <i>same event</i>)	1. Lexical Context: The sentence's Longformer Embedding. 2. Narrative Context: Its Updated Paragraph Vector (from Step 2). 3. Event Context: The Mean of its own Event Embeddings.
A Cross-View Attention module lets the Factual and Interpretive views "talk" to each other, modeling how facts are framed	A standard GNN (GAT) processes this graph, allowing context from the <i>end</i> of the article to inform the <i>beginning</i>	These are concatenated and fed into a final classifier
Output: A summary vector for each paragraph	Output: A document-aware, context-rich vector for each paragraph	

Experimental Setup

Datasets:

Event Extractor Training: MAVEN-ERE

Primary Bias Evaluation: **BASIL** (89 articles, 3 outlets, ~8k sentences)

Protocol:

- Strict 10-Fold Article-Level Cross-Validation (prevents data leakage)

Baselines:

- a. **Longformer-only:** (No event graph)
- b. **Feature Concatenation** (Simple Graph)
- c. **Lei & Huang (2024):** (The current SOTA flat, single-view graph)

Main Results vs. Baselines

Our model significantly outperforms all baselines, validating our hypothesis

Model	Precision	Recall	F1
Baseline 2 - Feature Concat	0.435	0.455	0.445
Baseline 1 - Longformer Only	0.445	0.470	0.457
Ablation 1 - Unified Graph	0.448	0.495	0.470
Ablation 2 - Flat Graph (SOTA)	0.452	0.510	0.479
Ablation 3 - No Cross-View	0.455	0.520	0.485
Final - Dual View Full	0.460	0.540	0.497

Ablation Studies: Why Our Model Works

All components of our design prove to be critical for performance

Component Removed	P	R	F1	$\Delta F1$
Full Model (All Components)	46.0	54.0	49.7	baseline
Remove Cross-View Attention (Abl 3)	45.5	52.0	48.5	-1.2
Remove Hierarchy (Abl 2)	45.2	51.0	47.9	-1.8
Remove Dual-View (Abl 1)	44.8	49.5	47.0	-2.7
Remove All Graphs (Baseline 1)	44.5	47.0	45.7	-4.0

Key Insights

1. **Dual-View Separation = Largest Impact (+2.7 F1)**

- Splitting factual/interpretive events provides biggest single improvement, validates core hypothesis about epistemological distinction

2. **Two-Level Hierarchy = Second Largest (+1.8 F1)**

- Multi-scale processing captures both local and global bias patterns, paragraph + document levels are complementary

3. **Cross-View Attention = Final Boost (+1.2 F1)**

- Modeling fact-framing interaction adds meaningful improvement, bidirectional attention learns characterization patterns

4. **All Components Essential - Each ablation degrades performance**

- Total improvement: +4.0 F1 over Longformer baseline, progressive gains demonstrate architectural synergy

Performance Gap Analysis

Our Results vs Lei & Huang (2024)

Metric	Our Model	Lei & Huang (2024)	Gap
Precision	46.0%	50.1%	-4.1
Recall	54.0%	54.1%	-0.1
F1	49.7%	52.0%	-2.3

Key Finding: We Match Their Recall!

- Our recall (54.0%) $\approx$ Their recall (54.1%)
- This means we're equally good at detecting bias sentences
- The gap is in precision (more false positives)

Why the 2.3 F1 Point Gap?

1. Limited Event Extractor Training (20% MAVEN-ERE)

- We trained on: 583 articles (20% subset)
- They trained on: 2,913 articles (100% dataset)

2. Partial BASIL Evaluation (51% of dataset)

- We classified: 153 articles (API constraints: 14 req/min, 1000/day)
- They evaluated on: 300 articles (full dataset)
- Impact:
 - Fewer training examples per fold (~120-130 vs ~240)
 - Subset may have different difficulty distribution
 - Selection gap (triplets 0-46, 86-89) missing middle range

Qualitative Analysis - Controversy Framing Detection

Pattern: One Fact + Two Opposite Interpretations = Bias
Case 1: True Positive - Controversy Framing Detected

"The Senate passed the healthcare bill 52-48, with Democrats celebrating it as a historic achievement while Republicans condemned it as a reckless disaster."

Ground Truth: 1 (biased)

Model Performance	Event Extraction	F-I Analysis
Baseline: P(bias) = 0.44 ✗ (missed - below 0.5)	Factual: [passed, bill 52-48]	Ratio: 2 factual : 4 interpretive (67% interpretive) Polarity: 2 positive vs 2 negative (conflict present)
Full Model: P(bias) = 0.78 ✓ (correctly detected)	Interpretive (Positive): [celebrating, historic achievement] Interpretive (Negative): [condemned, reckless disaster]	Structure: Single fact + opposite characterizations IN SAME SENTENCE

Why Is this Sentence Biased?

Controversy framing: Emphasizes conflict ("celebrating" vs "condemned")

Extreme language: "historic achievement" vs "reckless disaster"

False dichotomy: Only two interpretations (no moderate views)

Drama over substance: Focuses on reactions, not policy details

Neutral Alternative: "The Senate passed the healthcare bill 52-48."

Why Did the Baseline Miss? (0.44)

Sees both positive and negative words → they "cancel out"

Cannot separate facts from interpretations

Treats "historic" and "reckless" as competing signals that neutralize

Mixed polarity confuses single-view model

How Our Dual-View Detected The Bias (0.78)

Factual Subgraph

[passed] → [52-48]

Clean factual structure

Interpretive Subgraph

Positive: [celebrating] → [historic achievement]

Negative: [condemned] → [reckless disaster]

Two opposing opinion chains

Cross-View Attention

[passed] ← [historic achievement]
(positive)

[passed] ← [reckless disaster] (negative)

Both attach to same fact

Detection Signal

- Opposite polarities on same anchor
- Controversy framing pattern
- Contrastive learning recognizes this

Result

$P(\text{bias}) = 0.78 \checkmark$

Baseline = 0.44 (missed - mixed signals cancel)

Dual-view separates fact from framings

Case 2: Another Controversy Framing Example

Sentence (S2):

"Officials announced the tax reform plan, which supporters call essential for growth but critics describe as a giveaway to the wealthy."

Ground Truth: 1 (biased)

Model Performance	Event Extraction	F-I Analysis:
Baseline: P(bias) = 0.52 ✓ (barely detected)	Factual: [announced, tax reform plan]	Ratio: 2 factual : 2 interpretive (50% interpretive)
Full Model: P(bias) = 0.71 ✓ (strongly detected)	Interpretive (Negative): [giveaway to the wealthy]	Polarity: 1 positive vs 1 negative (direct opposition)
	Interpretive (Positive): [essential for growth]	Controversy framing: Plan presented as divisive

Why This is Biased:

The Fact	The Bias
• Tax reform exists • It was announced • (These are neutral, verifiable)	• "Essential for growth" (positive characterization) • "Giveaway to wealthy" (negative characterization) • Presenting BOTH → frames policy as controversial

Case 3: Pure Factual (No Interpretations) - Correctly Classified Neutral

Sentence (S3):

"The Senate voted 52-48 to approve the spending package on Tuesday."

Ground Truth: 0 (neutral)

Model Performance	Event Extraction	F-I Analysis
Baseline: P(bias) = 0.12 ✓ Correct	Factual: [voted, approve, 52-48, Tuesday]	Ratio: 4 factual : 0 interpretive (0% interpretive)
Full Model: P(bias) = 0.09 ✓ Even more confident	Interpretive: [] (empty)	No cross-view edges (nothing to connect)
		Pure verification: Numbers, actions, dates only

Challenges We Faced



API Rate Limits (Gemini)

- 14 requests/minute, 1000/day
- Took 7-8 days to classify events
- **Impact:** Could only classify 153 of 300 BASIL articles (51%)



Computational Constraints

- MAVEN-ERE too large for full training
- Used 20% subset (583 vs 2,913 articles)
- **Impact:** Event extractor potentially weaker than possible



Model Collapse During Training

- Initial runs: model predicted all non-bias
- **Solution:** Conservative hyperparameters, weighted loss (1:3 ratio)
- Required multiple training attempts to stabilize



Time Constraints

- 9-stage pipeline took 6+ weeks
- Limited hyperparameter search
- Couldn't evaluate on BiasedSents (ran out of API quota)

Limitations and Future Work



Dataset Coverage

- BASIL: Only 51% classified (153/300 articles)
- Triplet gap: Missing middle range (47-85), may affect difficulty distribution
- Training examples: ~120-130 per fold vs ~240 if complete



Training Data

- MAVEN-ERE: Only 20% used (583/2,913 articles)
- Full training might reach 92-93% F1 (vs current 89.40%)
- Better extraction = less error propagation downstream



Generalization Uncertainty

- Evaluated only on U.S. political news (2010-2019)
- Unknown performance on other domains, cultures, time periods